nanochat-turk
Tokenizer study results

Research brief · 9 November 2026 · Consolidated on main

12 tokenizers. 11 matched base models. Three vocabulary tiers.
Raw byte-level BPE versus MorphBPE trained with TRmorph, Zemberek and TurkishDelightNLP boundaries.

What the evidence supports	Result
Morphology / compression	MorphBPE preserves more automatic TRmorph boundaries; raw BPE uses fewer tokens per document word.
Pretraining	Raw BPE has the lowest archived validation bits per byte in every vocabulary tier.
Downstream evaluation	The legacy table records a +1.04-point 32k Zemberek difference. Generation and likelihood bugs make tokenizer-effect rankings provisional until a corrected matched rerun.
Missing model	32k TurkishDelight has a tokenizer, but no complete model in the recovered ablation inventory.

Compare within each tier. The architecture is constant within a vocabulary tier. Across tiers, depth changes (d20 / d16 / d12) to keep parameters near 0.9B. All completed ablation runs use 17,100 steps and 17.93B token positions. Equal token positions do not guarantee equal raw-text exposure or evaluation spans.

The package preserves the earlier 50k-document metrics, recovers a completed 93.21M-document training-corpus pass, and adds fresh matched train/validation measurements. Morphology references are automatic, not human gold annotation. Model differences are single-run estimates without confidence intervals.

New tokenizer measurements / validation split

50,000 documents; 500,000 isolated word occurrences per tokenizer. All rows share the same word/reference sample hashes. Percentages use occurrence counts.

Vocab	Tokenizer	Corpus tok/word	Isolated tok/word	One-token words %	Char-only words %	Morphemes intact %	Isolated boundary P %	Isolated boundary R %	Isolated boundary F1 %
32,768	Raw BPE	1.5728	2.0076	32.00	2.43	31.57	18.87	36.88	24.96
32,768	TRmorph	1.7785	1.9414	35.89	2.35	44.30	29.70	54.23	38.38
32,768	Zemberek	1.7673	1.9400	35.48	2.36	40.79	26.63	48.56	34.40
32,768	TurkishDelight	1.6989	1.9949	34.11	2.37	39.81	26.09	50.34	34.37
65,536	Raw BPE	1.4483	1.7850	41.48	2.31	37.15	21.99	33.48	26.54
65,536	TRmorph	1.6775	1.7121	45.67	2.25	50.27	35.88	49.55	41.62
65,536	Zemberek	1.6733	1.7176	44.74	2.25	46.39	31.77	44.21	36.97
65,536	TurkishDelight	1.5760	1.7695	43.09	2.25	45.27	30.84	46.02	36.93
131,072	Raw BPE	1.3611	1.5902	52.07	2.24	41.39	25.48	29.16	27.20
131,072	TRmorph	1.6001	1.5576	53.45	2.22	53.55	41.69	45.09	43.33
131,072	Zemberek	1.5989	1.5639	52.42	2.23	49.83	36.78	40.23	38.43
131,072	TurkishDelight	1.4809	1.6022	51.07	2.22	48.26	35.52	41.49	38.27

Morpheme preservation: fraction of reference morpheme spans that coincide exactly with one token span. Boundary consistency: micro precision/recall/F1 over isolated-word internal token and reference boundaries; these differ from contextual document-boundary preservation. Character-only: every token is one Unicode character; UTF-8 byte fragments are treated separately in JSON.

All twelve tokenizers are byte-complete and have no UNK mechanism, so UNK counts are N/A. Whole-word coverage is measured in isolation. Corpus fertility includes punctuation/spacing tokens. Automatic TRmorph references may favor TRmorph-trained tokenizers. Consult the JSON for alignment exclusions, denominators, byte-only coverage and roundtrip counts.

New tokenizer measurements / matched training sample

50,000 documents; 500,000 isolated word occurrences per tokenizer. All rows share the same word/reference sample hashes. Percentages use occurrence counts.

Vocab	Tokenizer	Corpus tok/word	Isolated tok/word	One-token words %	Char-only words %	Morphemes intact %	Isolated boundary P %	Isolated boundary R %	Isolated boundary F1 %
32,768	Raw BPE	1.6157	2.0472	31.05	2.31	30.76	19.08	37.17	25.22
32,768	TRmorph	1.8166	1.9890	34.09	2.23	42.90	29.36	54.02	38.05
32,768	Zemberek	1.7986	1.9810	34.06	2.23	39.60	26.54	48.44	34.29
32,768	TurkishDelight	1.7366	2.0357	32.73	2.25	38.46	25.94	49.98	34.15
65,536	Raw BPE	1.4856	1.8322	39.44	2.21	35.59	21.85	33.83	26.55
65,536	TRmorph	1.7121	1.7625	43.17	2.14	48.16	34.84	49.42	40.87
65,536	Zemberek	1.7030	1.7616	42.78	2.14	44.50	31.11	44.08	36.48
65,536	TurkishDelight	1.6091	1.8192	40.76	2.15	43.24	30.12	45.90	36.37
131,072	Raw BPE	1.3915	1.6372	49.58	2.04	39.47	24.92	29.55	27.04
131,072	TRmorph	1.6304	1.6027	50.73	2.02	51.33	40.14	45.00	42.43
131,072	Zemberek	1.6248	1.6026	50.17	2.02	47.77	35.67	39.99	37.71
131,072	TurkishDelight	1.5109	1.6481	48.56	2.02	46.27	34.58	41.69	37.80

Morpheme preservation: fraction of reference morpheme spans that coincide exactly with one token span. Boundary consistency: micro precision/recall/F1 over isolated-word internal token and reference boundaries; these differ from contextual document-boundary preservation. Character-only: every token is one Unicode character; UTF-8 byte fragments are treated separately in JSON.

All twelve tokenizers are byte-complete and have no UNK mechanism, so UNK counts are N/A. Whole-word coverage is measured in isolation. Corpus fertility includes punctuation/spacing tokens. Automatic TRmorph references may favor TRmorph-trained tokenizers. Consult the JSON for alignment exclusions, denominators, byte-only coverage and roundtrip counts.

Earlier metrics / recovered complete training corpus

Job 496882: COMPLETED, exit 0:0. All 12 tokenizers: 93,211,827 documents, 36.70B regex words, 299.17GB raw UTF-8 bytes. Isolated-word/alignment metrics use 500k occurrences; consistency uses 50k unique words.

Vocab	Tokenizer	Tok/word	Bytes/token	Morph edit	Exact morph sequence %	Boundaries preserved %	Morph consistency F1
32,768	Raw BPE	1.5553	5.2415	1.7096	32.46	14.66	0.3241
32,768	TRmorph	1.7698	4.6061	1.4319	41.54	54.18	0.5129
32,768	Zemberek	1.7551	4.6446	1.4986	39.50	40.44	0.4357
32,768	TurkishDelight	1.6865	4.8335	1.5544	37.54	34.43	0.4795
65,536	Raw BPE	1.4328	5.6894	1.4627	40.04	9.34	0.2863
65,536	TRmorph	1.6712	4.8780	1.1697	50.56	50.59	0.4421
65,536	Zemberek	1.6640	4.8989	1.2527	47.82	36.97	0.3743
65,536	TurkishDelight	1.5651	5.2084	1.3071	45.56	27.03	0.4296
131,072	Raw BPE	1.3473	6.0506	1.2623	47.01	5.51	0.2412
131,072	TRmorph	1.5952	5.1101	1.0045	56.66	46.32	0.3786
131,072	Zemberek	1.5916	5.1219	1.0859	53.80	33.63	0.3330
131,072	TurkishDelight	1.4715	5.5399	1.1345	51.37	20.64	0.3971

All twelve record zero document roundtrip failures. Preserved boundaries = 1 - crossed-reference-boundary rate; it does not penalize extra splits and differs from boundary precision/F1. The original morphology-consistency statistic uses shared-token/shared-morpheme word pairs. It is distinct from the new literal boundary metric.

This is a training-corpus diagnostic. The earlier 50k-document table, external Kumru/Cosmos tokenizer references, all distributions, timing data and original source files remain available in the companion repository report.

Pretraining and legacy CETVEL outcomes

Within-tier controls: same documented data source/order, architecture, optimizer recipe and 17.93B token positions. The historical recipe is retained in the pre-consolidation snapshot; original per-run Git revisions were not recorded. The consolidated d32 optimizer is a later recipe. Cross-tier comparisons change model depth.

Tier	Tokenizer	Validation BPB ↓	Token loss*	Core-11 ↑	Delta pp	CETVEL ex/s
32k/d20	Raw BPE	0.6232	2.4899	0.4514	+0.00	13.06
32k/d20	TRmorph	0.6266	2.0106	0.4541	+0.27	12.49
32k/d20	Zemberek	0.6250	2.3227	0.4618	+1.04	13.02
64k/d16	Raw BPE	0.6409	2.5812	0.4590	+0.00	15.13
64k/d16	TRmorph	0.6521	2.2754	0.4532	-0.58	15.23
64k/d16	Zemberek	0.6514	2.3013	0.4568	-0.22	15.38
64k/d16	TurkishDelight	0.6510	2.4869	0.4567	-0.23	16.09
128k/d12	Raw BPE	0.6749	3.0976	0.4651	+0.00	18.67
128k/d12	TRmorph	0.6917	2.5947	0.4503	-1.48	18.70
128k/d12	Zemberek	0.6940	2.6000	0.4618	-0.33	18.61
128k/d12	TurkishDelight	0.6820	2.6477	0.4481	-1.70	19.67

* Token loss uses different prediction units across tokenizers and is not a comparable loss measure. BPB is byte-normalized, but equal evaluation token budgets need not cover identical raw-text spans. Core-11 is the unweighted mean of eleven classification/likelihood tasks; its legacy values may be affected by continuation-boundary scoring. XQuAD is excluded and retained in the detailed legacy/fixed-harness reports. 32k TurkishDelight remains missing.

Later d32 production, continued d20 pretraining and changed benchmark harnesses are reported separately in the companion model report. They cannot be added as tokenizer-only treatments in this matched matrix.

Later training and corrected benchmark evidence

These recovered August results change model size, training budget or evaluator settings. They are separate from the eleven-model tokenizer ablation.

Model	Step	Token positions	Recorded BPB	Qualification
d20 BPE32k	34,200	35.861B	0.591451	Continued training; doubled budget
d32 BPE32k	28,000	58.720B	0.519040	Train-overlap diagnostic

The d32 model trained on all 30 source shards, including its validation shard. Its 0.519040 BPB cannot be interpreted as held-out generalization. This completed 2.819B-parameter FineWeb2 run is separate from the frozen WSD production family.

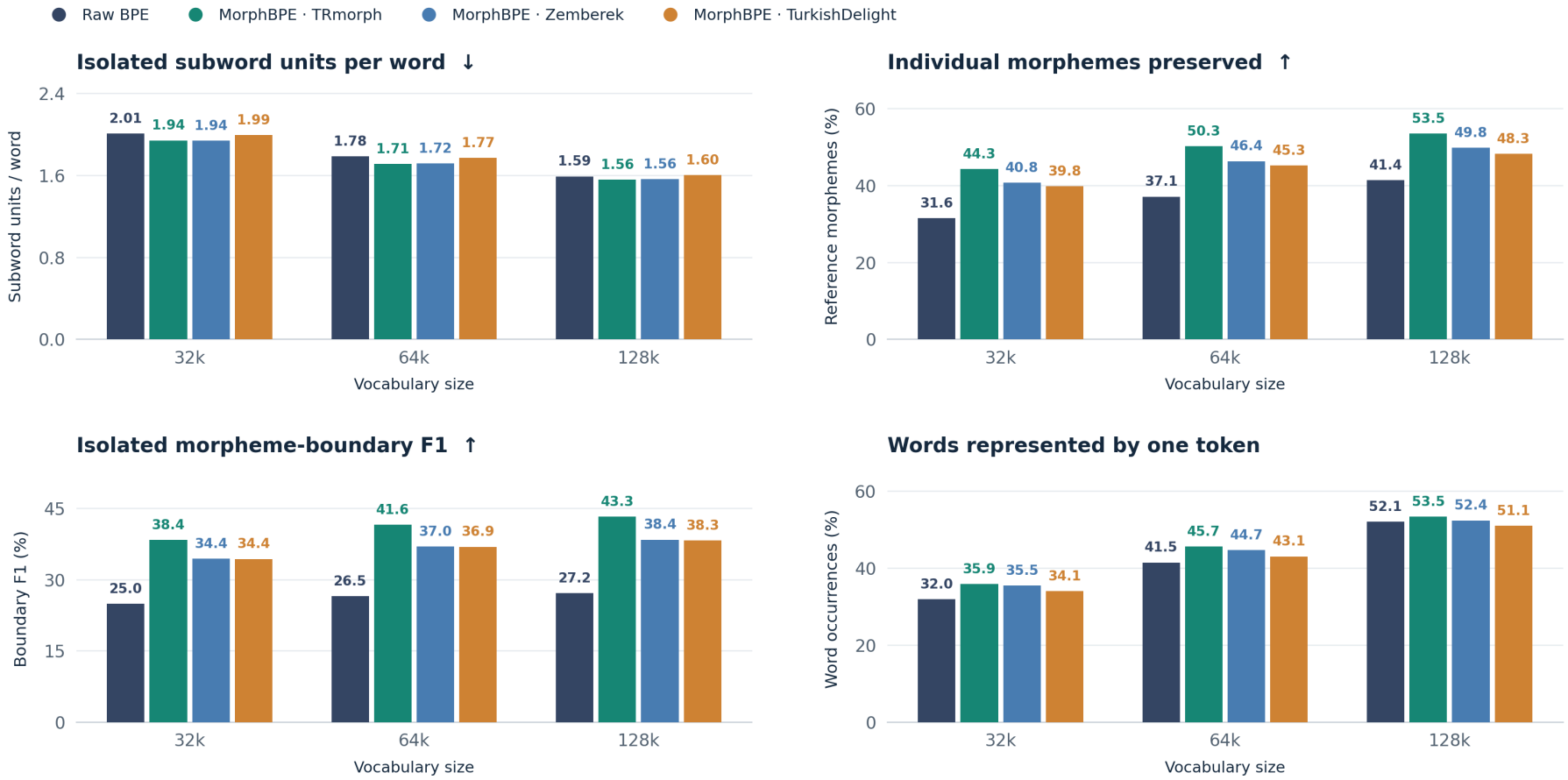
Model	XQuAD protocol	F1 /100	Exact match /100
d20 BPE32k, step 17100	June legacy adapter, cap 128	3.0985	0.0000
d20 BPE32k, step 17100	generation-fixed, CLI cap 128	11.3369	4.5378
d32 BPE32k, step 28000	generation-fixed, CLI cap 128	15.8833	8.2353
d32 BPE32k, step 28000	comparison runner, CLI cap 256	15.1717	8.2353

The d32 EACL table-compatible receipt reports 37.1709/100, the mean of seven category scores. It has its own scoring conventions and cannot be compared directly with core-11. Full task/category tables and protocol pins are in later_model_results.md.

The adapter now handles generation kwargs, stops and exact continuation boundaries, with regression tests and an output protocol version. A corrected common rerun for all eleven model cells is still needed; no tokenizer advantage is inferred from a single corrected baseline.

Twelve tokenizers on the same validation text

Fresh measurements · identical words within the sample · comparison within each vocabulary tier

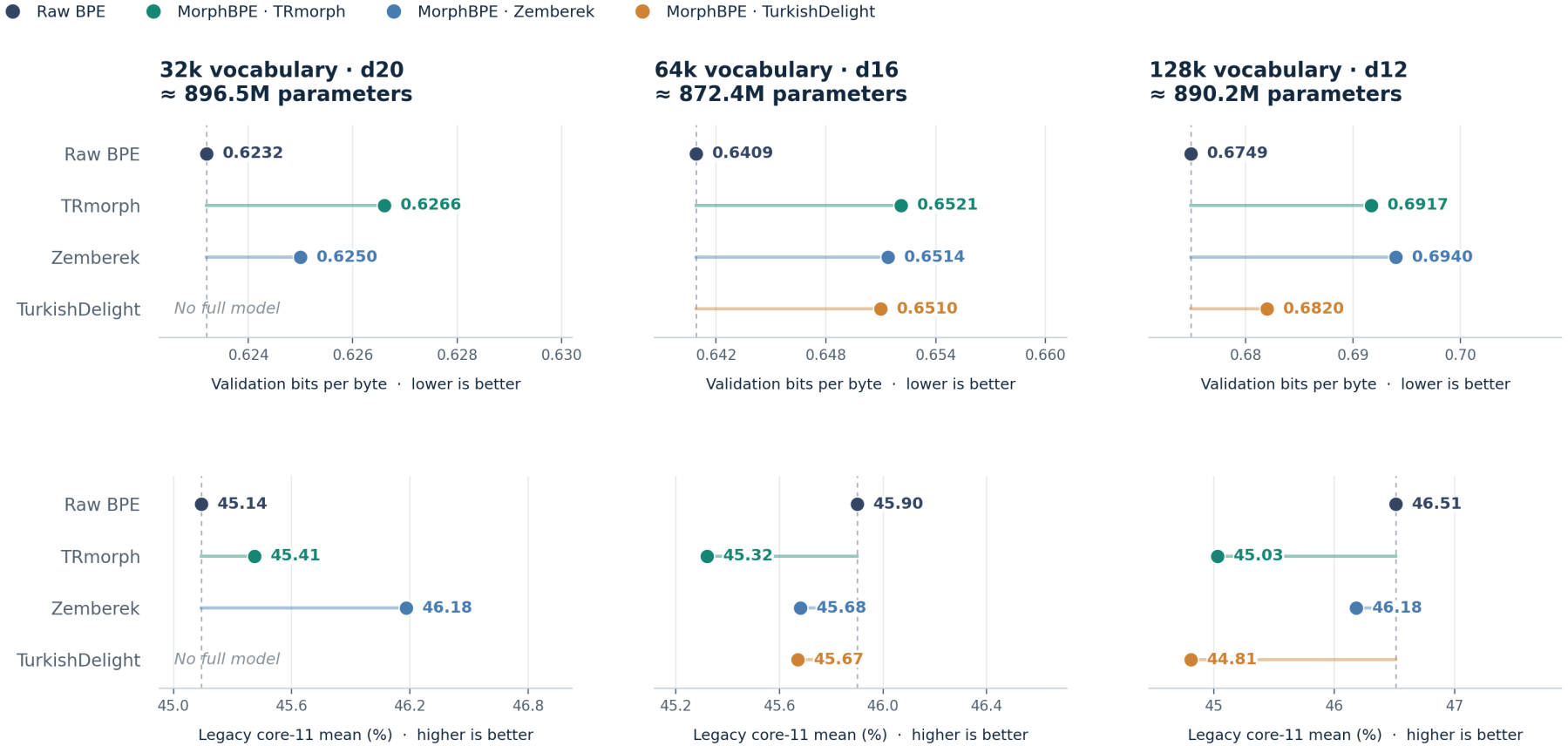


HELD-OUT FILE SPLIT · AUTOMATIC MORPHOLOGY REFERENCE

50,000 validation documents; 500,000 sampled word occurrences per tokenizer. Reference: automatic TRmorph, with no human annotation.
Morpheme preservation counts exact one-token reference spans; boundary F1 measures isolated-word internal boundary agreement.
TRmorph reference may favor TRmorph-trained tokenizers. Unalignable words are excluded from morphology denominators and reported separately.

Archived pretraining and legacy benchmarks

Raw BPE has lower archived BPB; core-11 rankings need a corrected common evaluator.

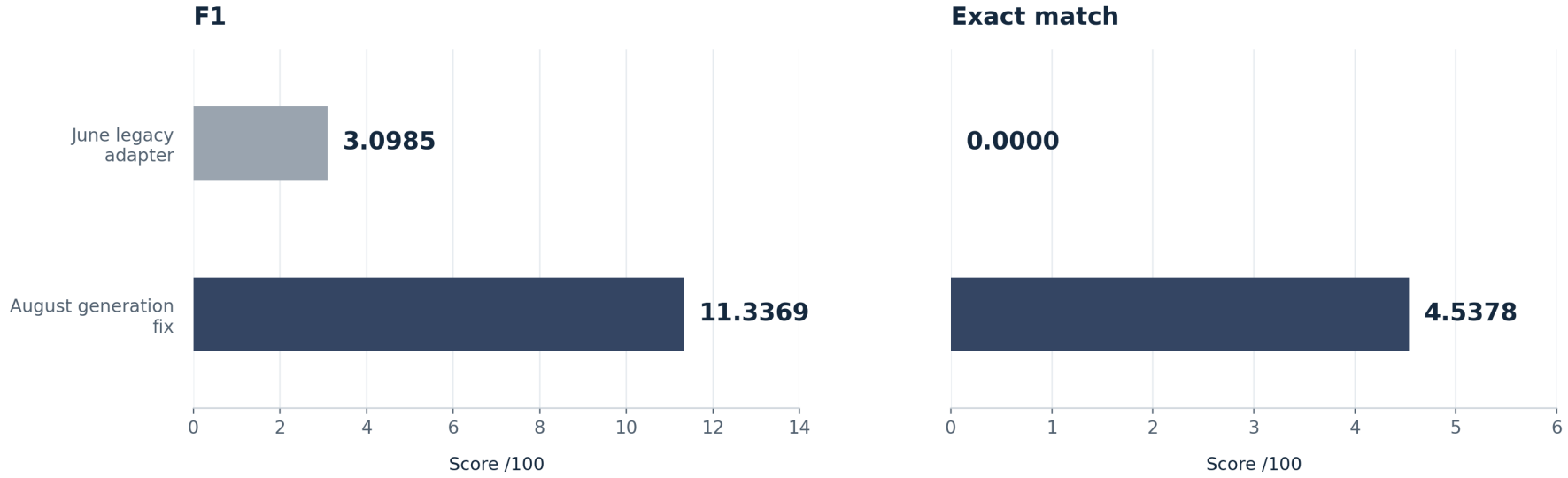


COMPARE WITHIN EACH COLUMN

Depth changes across tiers. All completed runs: 17,100 steps / 17.93B token positions; equal tokens do not imply equal raw-text exposure.
Legacy CETVEL has known continuation-boundary deviations; model rankings are provisional. Single runs; no paired uncertainty estimates.

An evaluator fix changes the same model’s score

d20 · raw BPE32k · checkpoint 17,100 · XQuAD · recorded CLI generation cap 128



THIS IS AN EVALUATOR CORRECTION, NOT A TOKENIZER OR TRAINING GAIN

The legacy generation adapter ignored task stop strings and generation limits. Corrected results for all MorphBPE models were not recovered.
Retain the original eleven-model table as legacy evidence; a common corrected evaluation is needed before claiming tokenizer effects.
Sources: archived June results and August reference-d20 fixed-generation run, recovered from UHeM on 9 Nov 2026.